

1. Which of the following is an example of Generative AI?

- (a) E-mail spam filter
- (b) ChatGPT writing an essay
- (c) Face unlock on a phone
- (d) Loan approval classifier

Answer: (b) ChatGPT writing an essay

2. What is a dataset?

- (a) A collection of related data
- (b) A computer program
- (c) A type of computer hardware
- (d) A programming language

Answer: (a) A collection of related data

3. In an 8-bit digital image, what is the valid range of numerical intensity values for a single pixel channel?

- (a) 0 to 1
- (b) 0 to 127
- (c) 0 to 255
- (d) -128 to 127

Answer: (c) 0 to 255

4. Which of the following is NOT a barrier to active listening?

- (a) Noise in the surroundings
- (b) Pre-judging the speaker
- (c) Maintaining eye contact
- (d) Thinking of your reply while the speaker is talking

Answer: (c) Maintaining eye contact

5. Which type of Data Mining is used to predict a continuous numerical value?

- (a) Classification
- (b) Regression
- (c) Clustering
- (d) Association

Answer: (b) Regression

6. Digital images are composed of tiny grid elements containing numerical values representing light intensity called:

- (a) Vectors
- (b) Pixels
- (c) Tensors
- (d) Neurons

Answer: (b) Pixels

7. The process of using domain knowledge to create new variables (features) that help machine learning algorithms work better is called _____.

Answer: Feature Engineering

8. Motivation that comes from within a person for personal satisfaction and joy is known as:
- (a) Extrinsic motivation
 - (b) Intrinsic motivation
 - (c) External incentive
 - (d) Reward-based motivation

Answer: (b) Intrinsic motivation

9. Which type of analytics **Answers** the question “What is likely to happen?”
- (a) Descriptive
 - (b) Diagnostic
 - (c) Predictive
 - (d) Prescriptive

Answer: (c) Predictive

10. Which Data Mining technique is mainly used to group similar data objects without predefined categories?
- (a) Classification
 - (b) Clustering
 - (c) Regression
 - (d) Prediction

Answer: (b) Clustering

11. In a confusion matrix, when the model predicts ‘Positive’ but the actual value is ‘Negative’, it is called a:
- (a) True Positive
 - (b) False Negative
 - (c) False Positive
 - (d) True Negative

Answer: (c) False Positive

12. Fill in the blank: In the sentence “Ravi quickly finished his Python assignment”, the word ‘quickly’ is a/an _____.

Answer: Adverb

13. Which programming language is most widely used to develop AI applications
- (a) Java
 - (b) Python
 - (c) C
 - (d) C++

Answer: (b) Python

14. Which stage of Data Science Methodology involves gathering user input on the deployed model's performance to refine future iterations?
- (a) Deployment
 - (b) Evaluation
 - (c) Feedback
 - (d) Modeling

Answer: (c) Feedback

15. What is the role of patterns in Data Mining?

- (a) They help identify useful relationships and trends in data.
- (b) They increase the size of a computer.
- (c) They improve the physical appearance of data.
- (d) They replace the need for data collection.

Answer: (a) They help identify useful relationships and trends in data.

16. In an 8-bit grayscale image, each pixel value ranges from _____ to _____.

Answer: 0 to 255

17. Which process deals with cleaning missing values, removing duplicates, and transforming variables before model training?

- (a) Data Collection
- (b) Data Preparation
- (c) Data Requirements
- (d) Data Evaluation

Answer: (b) Data Preparation

18. An e-mail has fixed fields (To, From, Subject, Date) but a free-text body. It is an example of:

- (a) Structured data
- (b) Semi-structured data
- (c) Unstructured data
- (d) Metadata only

Answer: (b) Semi-structured data

19. Which technique is used to assign data into already defined categories?

- (a) Clustering
- (b) Classification
- (c) Association
- (d) Data Collection

Answer: (b) Classification

20. What are metrics ? Why do we use it in the concepts of models.

Answer: Metrics are measures used to evaluate the performance of models. They help us understand how well a model performs.

21. The smallest unit of a digital image is called a _____.

Answer: Pixel

22. Why is the environmental impact considered a downside or limitation of modern AI?

- (a) Running large AI data centres requires massive amounts of electricity and water for cooling
- (b) AI code physically destroys the plastic casing of modern computers
- (c) Using AI prevents renewable solar panels from generating energy
- (d) AI programs release harmful toxic fumes directly into the user's room

Answer: (a) Running large AI data centres requires massive amounts of electricity and water for cooling

23. Which of the following is an important step before applying Data Mining techniques to a dataset?

- (a) Data Cleaning
- (b) Data Printing
- (c) Data Drawing
- (d) Data Typing

Answer: (a) Data Cleaning

24. State True or False: Feedback is the last stage of the Data Science Methodology.

Answer: True

25. What evaluation metric represents the ratio of True Positives to the total predicted positives

- (a) Recall
- (b) Precision
- (c) Accuracy
- (d) Specificity

Answer: (b) Precision

26. The observation that the number of transistors on a microchip doubles about every two years is known as _____.

Answer: Moore's Law

27. What is the main purpose of Data Mining?

- (a) To delete data
- (b) To discover useful patterns in data
- (c) To store files
- (d) To format documents

Answer: (b) To discover useful patterns in data

28. Which component of a neural network specifies how the weights and biases are adjusted during training?

- (a) Bias
- (b) Learning rule
- (c) Input layer
- (d) Output layer

Answer: (b) Learning rule

29. What self-management skill refers to the ability to work independently, stay focused, and take initiative without constant supervision?

- a) Self-awareness
- b) Self-motivation
- c) Stress management
- d) Self-regulation

Answer: b) Self-motivation

30. Identifying multiple objects in an image and drawing a bounding box around each of them is called _____.

Answer: Object Detection

31. What is a key limitation of even the smartest Artificial Intelligence systems today?

- (a) They cannot store any pictures or numbers

- (b) They completely lack real human feelings, empathy, and common sense
- (c) They take much longer than humans to add up two numbers
- (d) They only work when connected to a physical typewriter

Answer: (b) They completely lack real human feelings, empathy, and common sense

32. Which of the following is commonly used in Data Mining?

- (a) Finding patterns and relationships
- (b) Drawing pictures
- (c) Playing videos
- (d) Editing Text

Answer: (a) Finding patterns and relationships

33. In Machine Learning, when a model performs exceptionally well on training data but fails to generalize on unseen test data, it is said to be:

- a) Underfitting
- b) Overfitting
- c) Balanced
- d) Optimal

Answer: b) Overfitting

34. State True or False: Green jobs help reduce the consumption of energy and raw materials and minimise waste and pollution.

Answer: True

35. Which computer vision application involves identifying both what objects are present in an image and where they are located using bounding boxes?

- a) Image Classification
- b) Object Detection
- c) Image Compression
- d) Style Transfer

Answer: b) Object Detection

36. What is the main purpose of collecting data in research?

- (a) To increase the size of the project
- (b) To obtain information for analysis
- (c) To create charts only
- (d) To write a report

Answer: (b) To obtain information for analysis

37. In a confusion matrix, what does a "False Negative" (FN) represent?

- a) Actual positive predicted as positive
- b) Actual negative predicted as positive
- c) Actual positive predicted as negative
- d) Actual negative predicted as negative

Answer: c) Actual positive predicted as negative

38. The two neural networks that compete with each other in a Generative Adversarial Network (GAN) are the _____ and the _____.

Answer: Generator and Discriminator

39. Which primary Computer Vision process involves detecting specific structures like edges, corners, blobs, or textures within an image?

- a) Image Acquisition
- b) Pre-processing
- c) Feature Extraction
- d) Model Deployment

Answer: c) Feature Extraction

40. How is Data Mining useful in the field of computer science?

Answer: Data Mining is useful for discovering useful patterns, relationships, and trends in large amounts of data.

41. Which stage of the Data Science Methodology is generally the most time-consuming?

- (a) Data Collection
- (b) Data Preparation
- (c) Modelling
- (d) Deployment

Answer: (b) Data Preparation

42. Ananya practises AI coding every evening only because she enjoys solving puzzles, not for marks. This is an example of _____ motivation.

- (a) Extrinsic
- (b) Intrinsic
- (c) Negative
- (d) Financial

Answer: (b) Intrinsic

43. Which domain of AI gives computers "eyes" to recognize faces in photos or look at traffic on roads?

- a) Audio Recording
- b) Computer Vision
- c) Battery Charging
- d) Text Typing

Answer: b) Computer Vision

44. Which of the following is an example of Classification?

- (a) Predicting the price of a house
- (b) Grouping customers based on similarities
- (c) Classifying emails as spam or not spam
- (d) Finding products purchased together

Answer: (c) Classifying emails as spam or not spam

45. In Freytag's Pyramid, the stage at which the story reaches its turning point or highest tension is called the _____.

Answer: Climax

46. The technology that extracts written or printed text from images and translates it into machine-encoded text is known as:

- a) Facial Recognition

b) Optical Character Recognition (OCR)

c) Object Segmentation

d) Generative Adversarial Network

Answer: b) Optical Character Recognition (OCR)

47. A dataset contains thousands of customer records. Which Data Mining process would be most useful for discovering relationships between products frequently purchased together?

(a) Association

(b) Classification

(c) Image Recognition

(d) Data Entry

Answer: (a) Association

48. Define the F-1 Score metric.

Answer: F1-Score is the harmonic mean of Precision and Recall.

49. Which type of neural network is most commonly used for image recognition?

(a) Recurrent Neural Network

(b) Convolutional Neural Network

(c) Single Perceptron

(d) None of these

Answer: (b) Convolutional Neural Network